

Stabilizer states are theta series

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1 Qudits

An N qudit state over a finite field $\mathbf{F}$ is a vector in the projective Hilbert space

$$\mathbf{H} = \mathbf{C}[\mathbf{F}^N]/\mathbf{C}^*$$

where $\mathbf{C}$ is the field of complex numbers. Assume $\mathbf{F}$ has odd prime order p .

2 Pauli group

The Pauli group is a finite subgroup of the automorphism group of $\mathbf{H}$. It is generated by the linear operators Z_i, X_i defined by

$$Z_i|n\rangle = q^{n_i}|n\rangle \quad X_i|n\rangle = |n+e_i\rangle$$

for $i = 1, \dots, N$ and $n \in \mathbf{F}^N$, where $q = e^{2\pi i/p}$ is a p th root of unity and $e_i = (\delta_{ij})_j$ is the standard i th basis vector in $\mathbf{F}^N$. The generators have order p . They satisfy the commutation relations

$$Z_i Z_j = Z_j Z_i \quad X_i X_j = X_j X_i \quad Z_i X_j = q^{\delta_{ij}} X_j Z_i.$$

3 Stabilizer states

Stabilizer states are N qudit states fixed by N independent Pauli operators.

Theorem 1. Stabilizer states are theta series

$$\Theta = \sum_{n \in \mathbf{F}^N} |nB - uD\rangle q^{nAB^t n^t/2 - uCB^t n^t}$$

where $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is a symplectic matrix over $\mathbf{F}$ and u is a vector in $\mathbf{F}^N$.

Proof. We will show that $P_i\Theta = \Theta$ for Pauli operators

$$P_i = q^{u'_i} \prod_{j=1}^N Z_j^{A_{ij}} X_j^{B_{ij}}$$

for $i = 1, \dots, N$ where $u' = u - \text{diag}(AB^t/2)$.

Use the symplectic identities

$$AB^t = BA^t \quad DA^t - CB^t = I$$

to shift the summation variable n .

$$\begin{aligned} P_i\Theta &= \sum_n |(n+e_i)B-uD\rangle q^{(nAB^tn^t/2-uCB^tn^t)+(n+e_i)B-uD)A^te_i^t+(u-\text{diag}(AB^t/2))e_i^t} \\ &= \sum_n |(n+e_i)B-uD\rangle q^{nAB^tn^t/2+(e_iAB^t-uCB^t)n^t+(e_iAB^t/2+u-uDA^t)e_i^t} \\ &= \sum_n |(n+e_i)B-uD\rangle q^{nAB^tn^t/2+(e_iAB^t-uCB^t)n^t+(e_iAB^t/2-uCB^t)e_i^t} \\ &= \sum_n |(n+e_i)B-uD\rangle q^{(n+e_i)AB^t(n+e_i)^t/2-uCB^t(n+e_i)^t} \\ &= \Theta \end{aligned}$$

□

4 Clifford group

The Clifford group is a finite subgroup of the automorphism group of $\mathbf{H}$. It acts transitively on the set of stabilizer states. It contains the Pauli group and has three additional generators: the quadratic phase gate Q_i and the sum gate S_{ij} and the Fourier gate F_i defined by

$$Q_i|n\rangle = |n\rangle q^{n_i^2/2} \quad S_{ij}|n\rangle = |n(I+E_{ij})\rangle \quad F_i|n\rangle = \sum_{n'_i} |n(I-E_{ii})+n'_ie_i\rangle q^{n_in'_i}$$

where E_{ij} is the matrix with one at ij and zero everywhere else.

5 Symplectic representation

Collect the symplectic matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{F}^{(N+N) \times (N+N)}$ and the phase vector $u \in \mathbf{F}^N$ and any other vector $v \in \mathbf{F}^N$ into the affine symplectic matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ u^t & A & B \\ v^t & C & D \end{pmatrix}.$$

The affine symplectic matrices form a group $\widehat{\text{Sp}}$. In Theorem 1 we defined a map

$$\begin{aligned} f : \widehat{\text{Sp}} &\rightarrow \mathbf{H} \\ M &\mapsto \Theta_M. \end{aligned}$$

Write Th for the image of $\widehat{\text{Sp}}$ in $\mathbf{H}$. The pullback $P_g : \widehat{\text{Sp}} \rightarrow \widehat{\text{Sp}}$ of a Clifford gate g along the map f makes the diagram

$$\begin{array}{ccc} \widehat{\text{Sp}} & \xrightarrow{f} & \text{Th} \\ P_g \downarrow & & \downarrow g \\ \widehat{\text{Sp}} & \xrightarrow{f} & \text{Th} \end{array}$$

commute.

Theorem 2. Clifford gates pull back to matrices in $\widehat{\text{Sp}}$. They act on the right.

g	Z_i	X_i	Q_i	S_{ij}	F_i
P_g	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I & 0 \\ e_i^t & 0 & I \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ -e_i^t & I & 0 \\ 0 & 0 & I \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I & 0 \\ 0 & E_{ii} & I \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I - E_{ji} & 0 \\ 0 & 0 & I + E_{ij} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I - E_{ii} & -E_{ii} \\ 0 & E_{ii} & I - E_{ii} \end{pmatrix}$

Proof. Qudit states live in projective space, so it suffices to show that

$$g\Theta_M = K_g \cdot q^{L_g} \Theta_{MP_g}$$

for each Clifford gate g , for constants $K_g \in \mathbf{C}^*$ and $L_g \in \mathbf{F}$.

Z gate

$$\begin{aligned}
Z_i \Theta_M &= \sum_n |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t + (nB - uD)e_i^t} \\
&= \sum_n |(n - e_i D^t)B - uD\rangle q^{(n - e_i D^t)AB^t(n^t - De_i^t) / 2 - uCB^t(n^t - De_i^t) + e_i B^t(n^t - De_i^t) - uDe_i^t} \\
&= \sum_n |(n - e_i D^t)B - uD\rangle q^{nAB^t n^t / 2 + e_i(I - D^t A)B^t n^t - uCB^t n^t + L} \\
&= q^L \sum_n |(n - e_i D^t)B - uD\rangle q^{nAB^t n^t / 2 - e_i B^t CB^t n^t - uCB^t n^t} \\
&= q^L \sum_n |nB - (u + e_i B^t)D\rangle q^{nAB^t n^t / 2 - (u + e_i B^t)CB^t n^t} \\
&= q^L \Theta_{MP_{Z_i}}
\end{aligned}$$

X gate

$$\begin{aligned}
X_i \Theta_M &= \sum_n |nB - uD + e_i\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \\
&= \sum_n |(n - e_i C^t)B - (u - e_i A^t)D\rangle q^{(n - e_i C^t)AB^t(n^t - Ce_i^t) / 2 - (u - e_i A^t)CB^t(n^t - Ce_i^t) + L} \\
&= q^L \sum_n |(nB - (u - e_i A^t)D)\rangle q^{nAB^t n^t / 2 - (u - e_i A^t)CB^t n^t} \\
&= q^L \Theta_{MP_{X_i}}
\end{aligned}$$

Q gate

$$\begin{aligned}
Q_i \Theta_M &= \sum_n |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t + (nB - uD)E_{ii}(B^t n^t - D^t u^t) / 2} \\
&= \sum_n |nB - uD\rangle q^{n(A + BE_{ii})B^t n^t / 2 - u(C + DE_{ii})B^t n^t + L} \\
&= q^L \Theta_{MP_{Q_i}}
\end{aligned}$$

S gate

$$\begin{aligned}
S_{ij} \Theta_M &= \sum_n |(nB - uD)(I + E_{ij})\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \\
&= \sum_n |nB(I + E_{ij}) - uD(I + E_{ij})\rangle q^{n(A(I - E_{ji}))(B(I + E_{ij}))^t n^t / 2 - u(C(I - E_{ji}))(B(I + E_{ij}))^t n^t} \\
&= \Theta_{MP_{S_{ij}}}
\end{aligned}$$

F gate

If the vector $e_i B^t$ is zero, then B has a nontrivial kernel vector $w \in \mathbf{F}^N$. Use it to break up the sum.

$$\begin{aligned}
F_i \Theta_M &= \sum_{\substack{n \in \mathbf{F}^N \\ n'_i \in \mathbf{F}}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times q^{nAB^t n^t / 2 - uCB^t n^t + (nB - uD)e_i^t n'_i} \\
&= \sum_{\substack{n \in \mathbf{F}^N / w \\ n'_i \in \mathbf{F}}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times \sum_{m \in \mathbf{F}} q^{(n+mw)AB^t (n+mw)^t / 2 - uCB^t (n+mw)^t - uDe_i^t n'_i} \\
&= \sum_{n, n'_i} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times \sum_m q^{nAB^t n^t / 2 - uCB^t n^t - uDe_i^t n'_i} \\
&= p \cdot \sum_{n, n'_i} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times q^{nAB^t n^t / 2 - uCB^t n^t - uDe_i^t n'_i}
\end{aligned}$$

Change variables $n'_i \mapsto -e_i A^t (n^t + n'_i w^t) + e_i C^t u^t$.

$$\begin{aligned}
F_i \Theta_M &= p \cdot \sum_{\substack{n \in \mathbf{F}^N \\ n'_i \in \mathbf{F}}} |(n + n'_i w)(B(I - E_{ii}) - AE_{ii}) - u(D(I - E_{ii}) - CE_{ii})\rangle \\
&\quad \times q^{nAB^t n^t / 2 - uCB^t n^t + uDE_{ii}(A^t(n^t + n'_i w^t) - C^t u^t)} \\
&= p \cdot q^L \sum_n |(n(B(I - E_{ii}) - AE_{ii}) - u(D(I - E_{ii}) - CE_{ii}))\rangle \\
&\quad \times q^{nAB^t n^t / 2 - u(CB^t - DE_{ii}A^t)n^t} \\
&= \Theta_{MP_{F_i}}
\end{aligned}$$

If the vector $e_i B^t$ is not zero then there is a vector $w \in \mathbf{F}^N$ with $wB = -e_i$. Use it to split the summation variable $n \mapsto n + mw$. Restrict the new n to the hyperplane $nBe_i^t = e_i Be_i^t$ to avoid over-counting.

$$\begin{aligned}
F_i \Theta_M &= \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times \sum_{m \in \mathbf{F}} q^{(n+mw)AB^t(n+mw)^t/2 - uCB^t(n+mw)^t + ((n+mw)B - uD)e_i^t n'_i} \\
&= \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle q^{nAB^t n^t/2 - uCB^t n^t + n'_i(e_i B - uD)e_i^t} \\
&\quad \times \sum_m q^{-m^2 w A e_i^t/2 - m(n'_i + (nA - uC)e_i^t)}
\end{aligned}$$

Change variables $n'_i \mapsto -n'_i w A e_i^t - (nA - uC)e_i^t$.

$$\begin{aligned}
F_i \Theta_M &= \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(n + n'_i w)(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{nAB^t n^t/2 - uCB^t n^t - (n'_i)^2 e_i A^t w^t - e_i B E_{ii} A^t (n^t + n'_i w^t) + u D E_{ii} A^t (n^t + n'_i w^t) + L} \\
&\quad \times \sum_m q^{-m^2 w A e_i^t/2 + m n'_i w A e_i^t}
\end{aligned}$$

The inner sum is a quadratic Gauss sum. It is equal to $K \cdot q^{(n'_i)^2 e_i A^t w^t/2}$ for some constant $K \in \mathbf{C}^*$.

$$\begin{aligned}
F_i \Theta_M &= K \cdot q^L \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(n + n'_i w)(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{nAB^t n^t/2 - uCB^t n^t - (n'_i)^2 e_i A^t w^t/2 - e_i B E_{ii} A^t (n^t + n'_i w^t) + u D E_{ii} A^t (n^t + n'_i w^t)} \\
&= K \cdot q^L \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(n + n'_i w)(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{(n+n'_i w)(A(I - E_{ii})B^t - B E_i A^t)(n^t + n'_i w^t)/2 - u(C(I - E_{ii})B^t - D E_{ii} A^t)(n^t + n'_i w^t)} \\
&= K \cdot q^L \sum_n |n(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{n(A(I - E_{ii}) + B E_{ii})((I - E_{ii})B^t - E_i A^t)n^t/2 - u(C(I - E_{ii}) + D E_{ii})((I - E_{ii})B^t - E_i A^t)n^t} \\
&= \Theta_{MP_{F_i}}
\end{aligned}$$

□

6 Example

The Clifford circuit

$$Z_1^3 Q_1^4 S_{12} F_1 X_2 |00\rangle$$

generates the twisted EPR pair

$$\Psi = \sum_{n \in \mathbf{F}} |n \ n+1\rangle q^{2n^2+3n}.$$

The symplectic representation is the matrix $P_{X_2} P_{F_1} P_{S_{12}} P_{Q_1^4} P_{Z_1^3} \in \widehat{\text{Sp}}$ defined by the product

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 4 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

which is equal to

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -3 & -4 & 0 & -1 & -1 \\ -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The phase vector and the symplectic blocks are

$$u = (-3 \ -1) \quad A = \begin{pmatrix} -4 & 0 \\ -1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The theta series is

$$\begin{aligned} \Theta &= \sum_{n \in \mathbf{F}^2} |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \\ &= \sum_{n_1, n_2} |(n_1 \ n_2) \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} - (-3 \ -1) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\rangle \\ &\quad \times q^{(n_1 \ n_2) \begin{pmatrix} -4 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} / 2 - (-3 \ -1) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}} \\ &= \sum_{n_1, n_2} |-n_1 \ 1 - n_1\rangle q^{2n_1^2 - 3n_1} \\ &= p \cdot \Psi. \end{aligned}$$